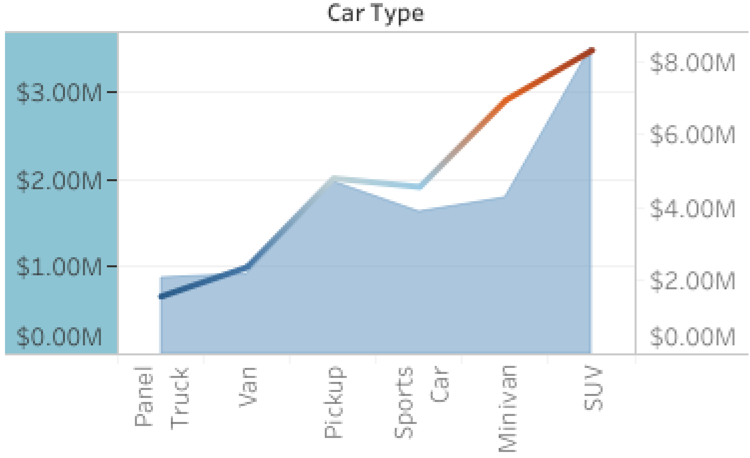
Data Vizualization Using Tableau

Objective of the Project:

This project aims at analyzing the car insurance claims data for finding the important trends and patterns and risk factors influencing incidence and amounts of the claim. The project uses data visualization and data exploration for gaining insights that can be used by insurance industries to improve their policy offerings, segmentation of customers, fraud detection, and optimization of premium strategies. Moreover, this research provides recommendations for risk minimization, better claims processes, and business sustainability along with customer sati..

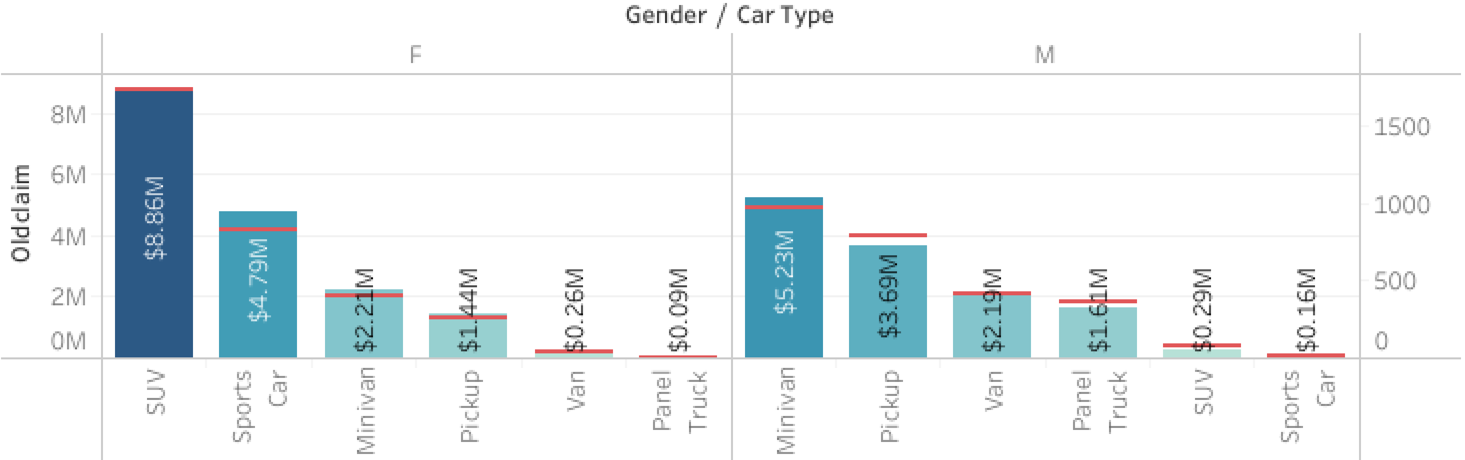
By
~Arindam Saha

Car Type VS Claims

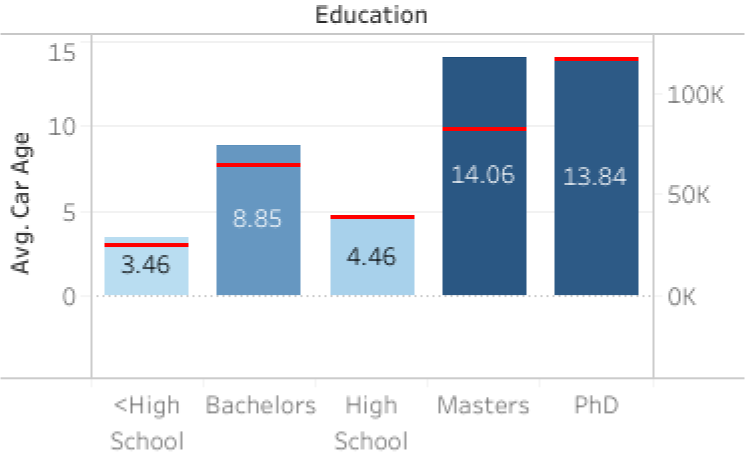


Gender vs OldClaim

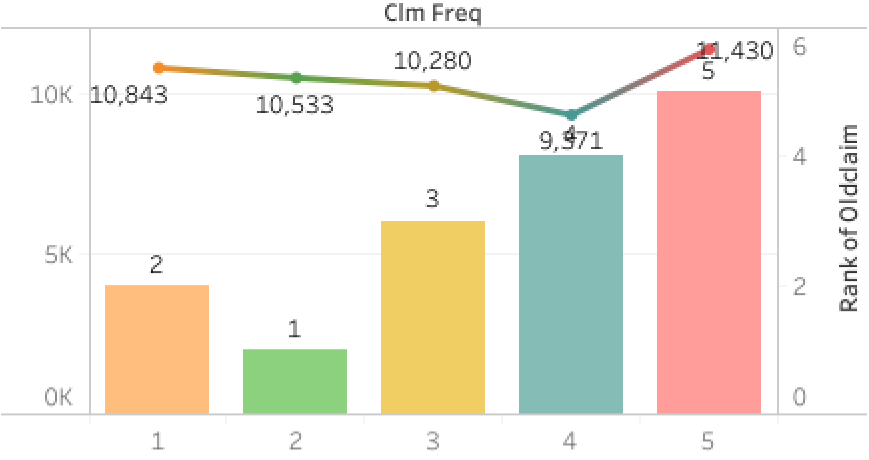
INSIGHTS



Edu VS income and car age



Claim Freq VS Old Claim

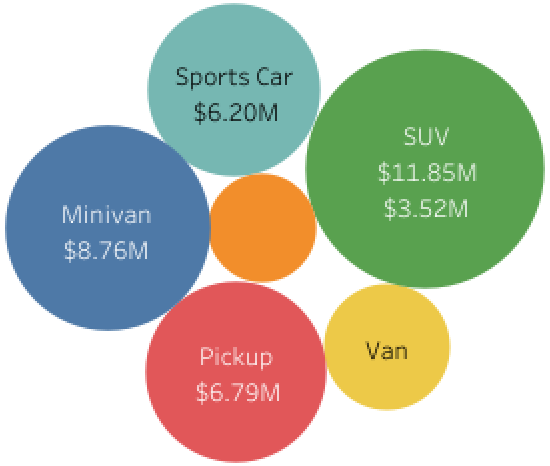


Gender vs Car Crash Claim

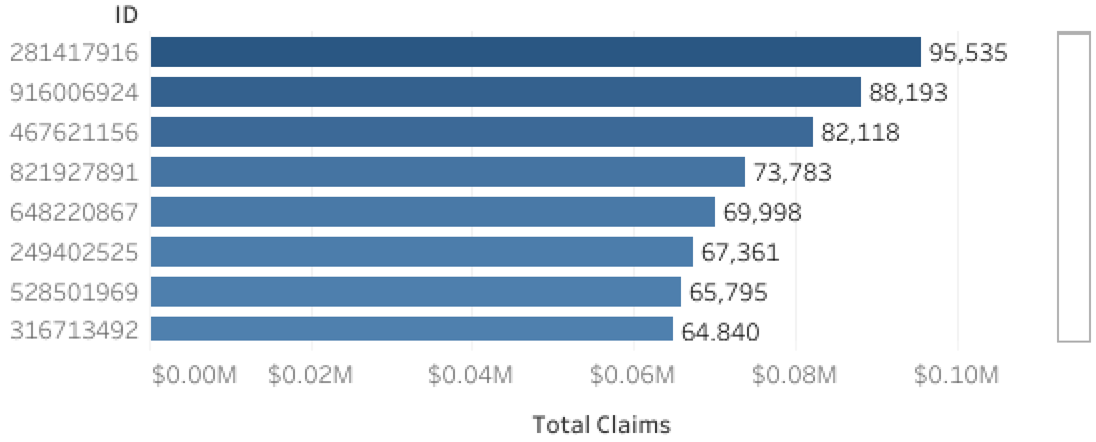
	Gender	
Car Type	F	M
Minivan	$0.57M	$1.30M
Panel Truck	$0.04M	$0.89M
Pickup	$0.52M	$1.57M
Sports Car	$1.65M	$0.05M
SUV	$3.52M	$0.18M
Van	$0.11M	$0.93M

1. Introduction | 2. Claims Analysis by Demographics | 3. Top Claimers and Yearly Claim Analysis | 4. Age-Based Claim Patterns | 5. Vulnerable Demographics in Clai.. | 6. Customer Profile | 7. Summary / Recommendations

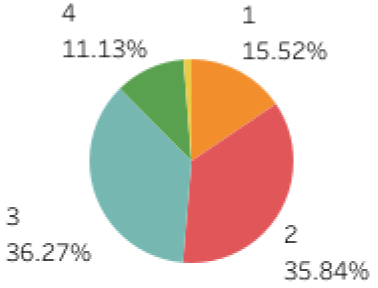
Car Type vs Total Claim



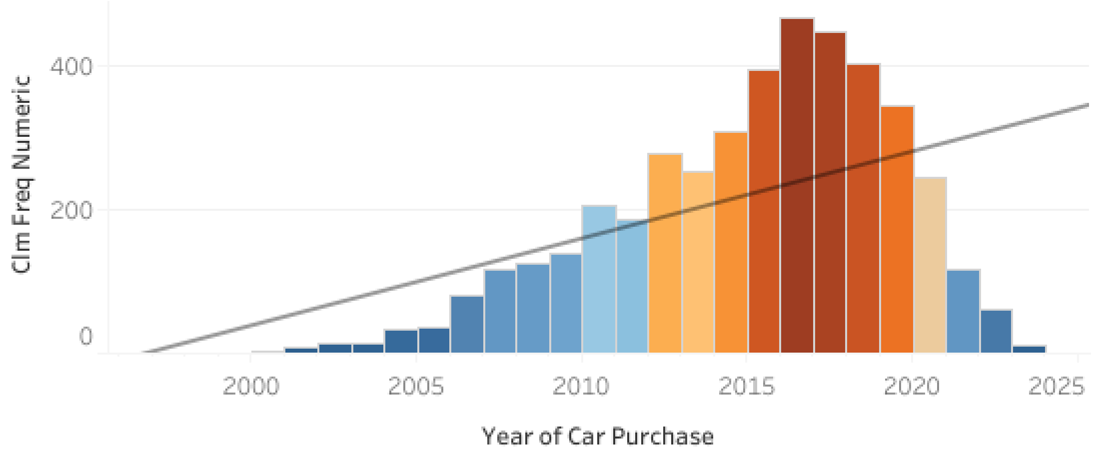
Top Claimers



Proportion of Claims by Frequency

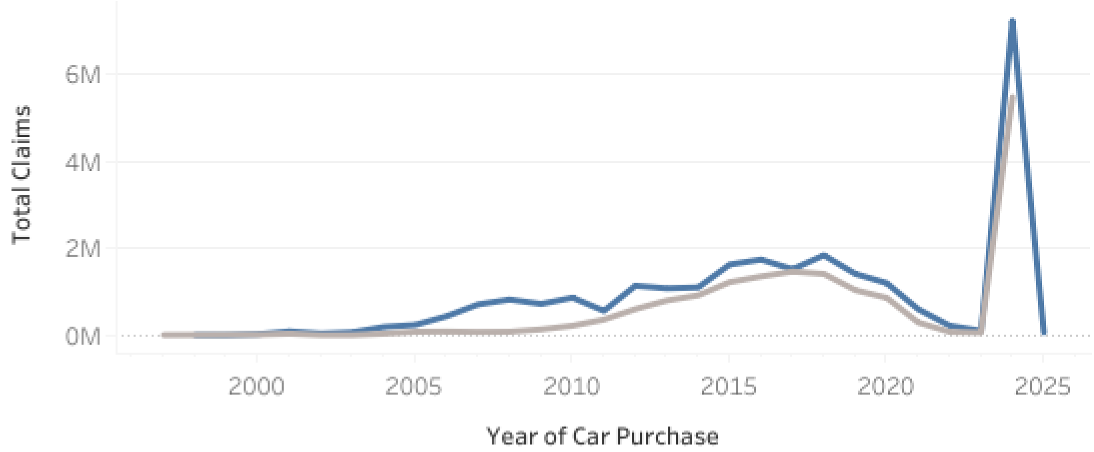


Avg Claim Freq by Year of Purchase



Total Claims by Year of Purchase

INSIGHTS



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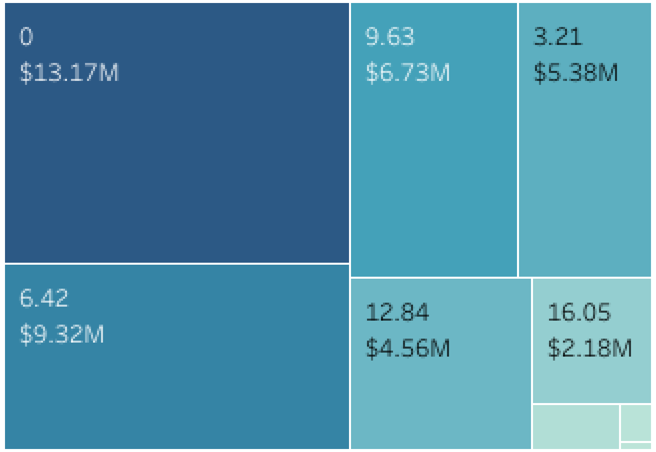
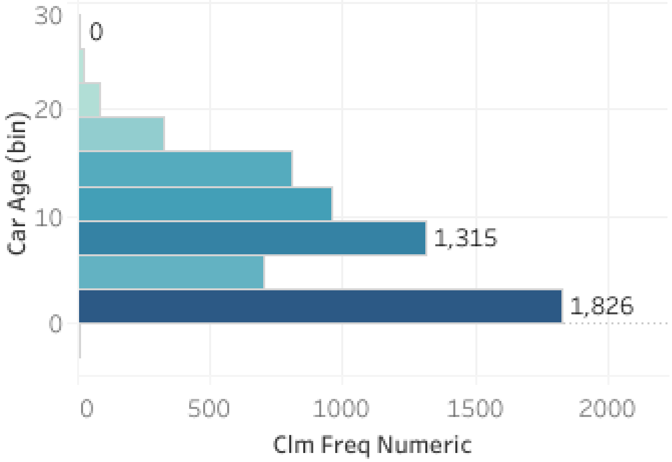
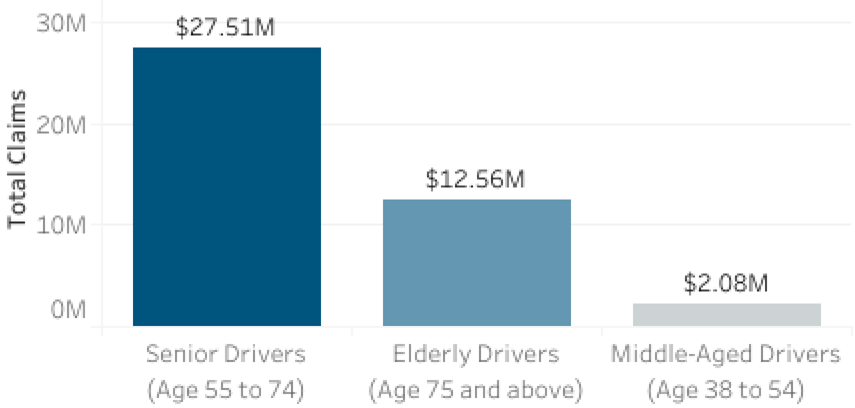
Car Age vs Total Claim



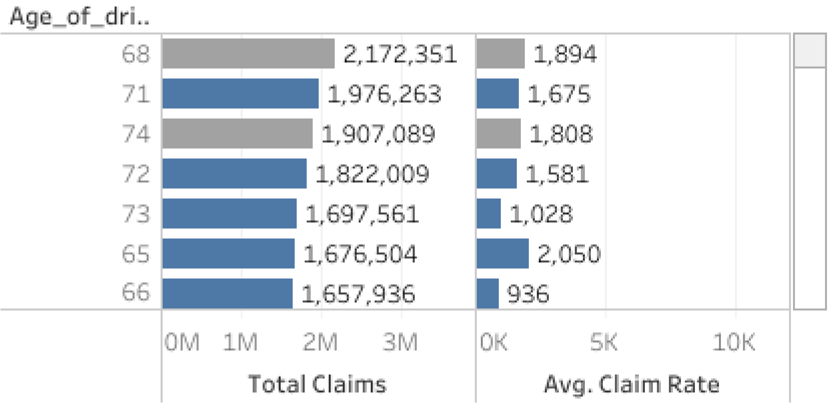
Car age vs Claim Freq



Driver Age Group vs Total Claim Dristribution

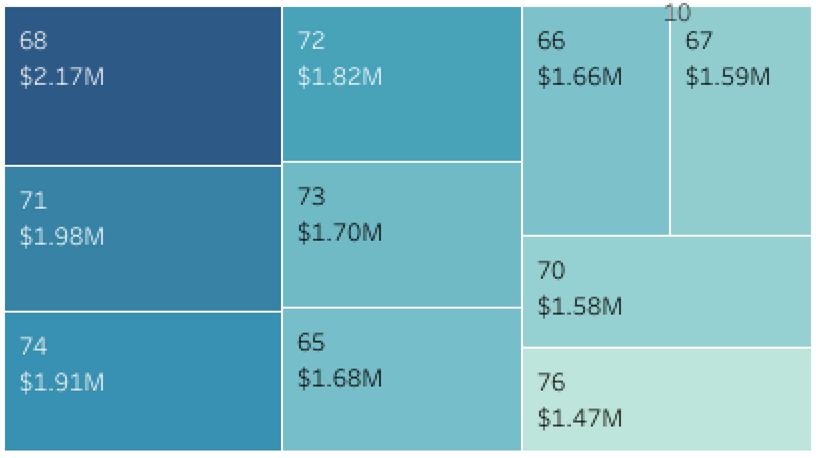


Top Driver Ages wrt AVG Claim Rate / Total Claim



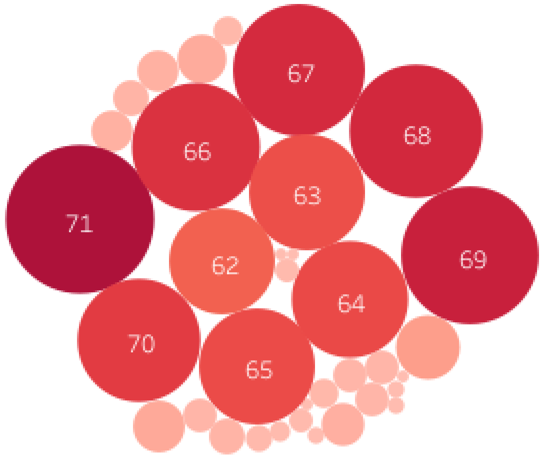
Top driver age towards total claim

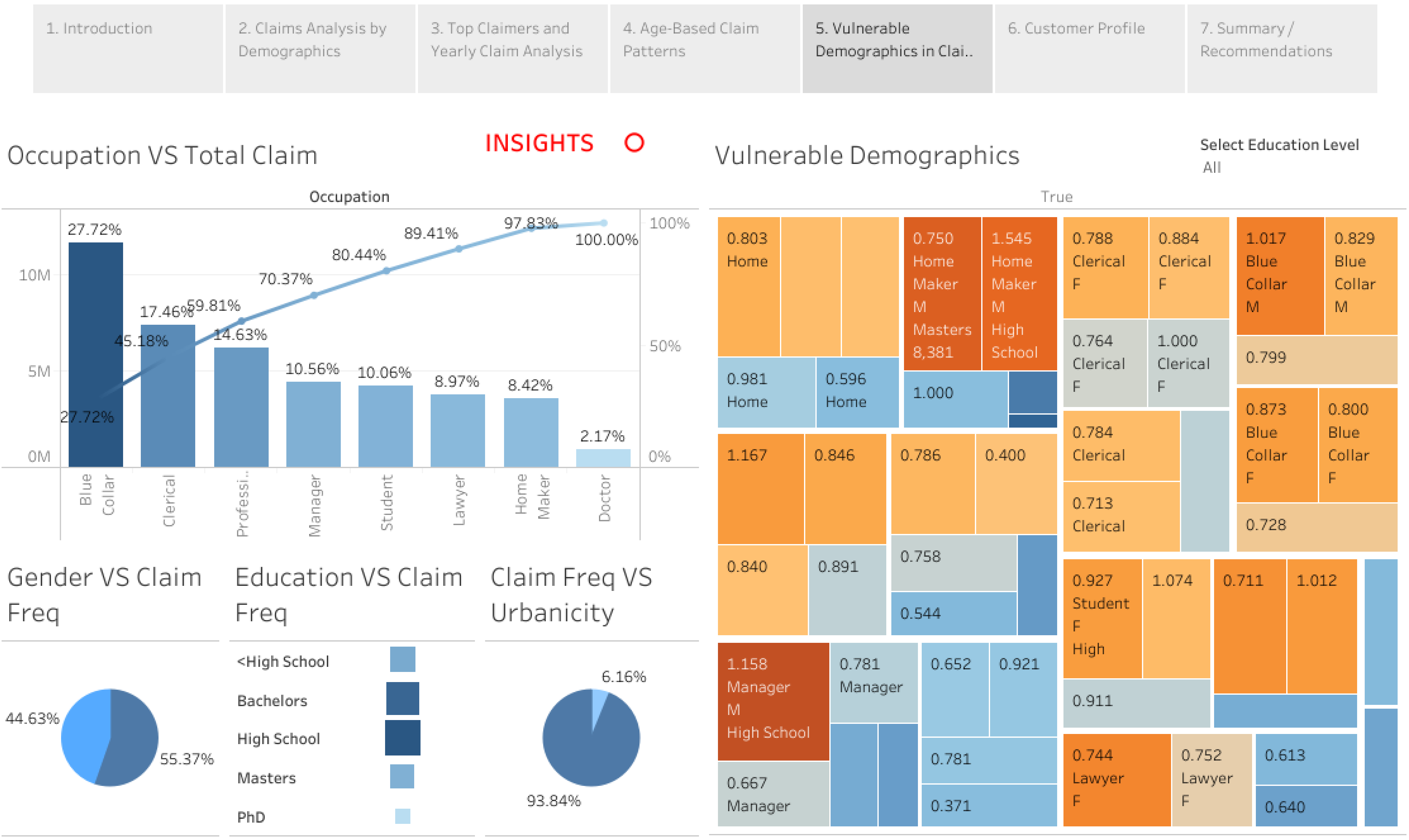
Top N Drive..
10



Kidsdrive

INSIGHTS

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Occupation VS Total Claim
INSIGHTS
Occupation
10M
5M
0M
100%
50%
0%
27.72%
17.46%
14.63%
10.56%
10.06%
8.97%
8.42%
2.17%
45.18%
59.81%
70.37%
80.44%
89.41%
97.83%
100.00%
Blue Collar
Clerical
Professi..
Manager
Student
Lawyer
Home Maker
Doctor
Vulnerable Demographics
Select Education Level
All
True
0.803 Home
0.750 Home Maker M Masters 8,381
1.545 Home Maker M High School
0.788 Clerical F
0.884 Clerical F
1.017 Blue Collar M
0.829 Blue Collar M
0.981 Home
0.596 Home
1.000
0.764 Clerical F
1.000 Clerical F
0.799
0.873 Blue Collar F
0.800 Blue Collar F
1.167
0.846
0.786
0.400
0.784 Clerical
0.713 Clerical
0.728
0.840
0.891
0.758
0.544
0.927 Student F High
1.074
0.711
1.012
0.911
1.158 Manager M High School
0.781 Manager
0.652
0.921
0.781
0.371
0.667 Manager
0.744 Lawyer F
0.752 Lawyer F
0.613
0.640
Gender VS Claim Freq
44.63%
55.37%
Education VS Claim Freq
<High School
Bachelors
High School
Masters
PhD
Claim Freq VS Urbanicity
6.16%
93.84%

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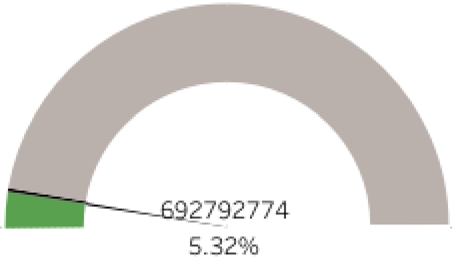
Back Button

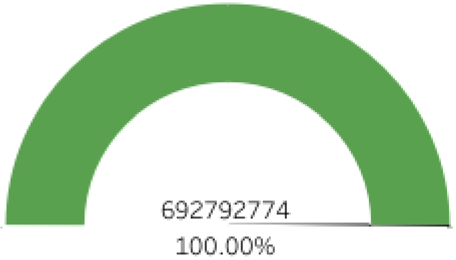
Car Age	Gender	Kids Drive	Driver Age	M Status	Education
1	F	0	81	Yes	High School

Income_



$19,532

TravelTime



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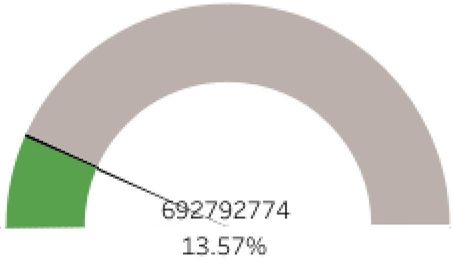
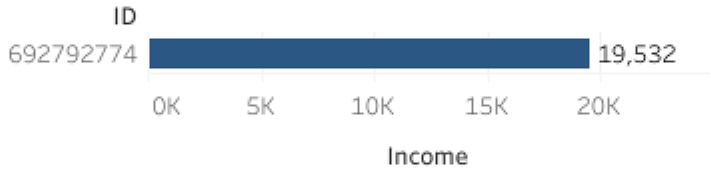
INSIGHTS

Blue Book

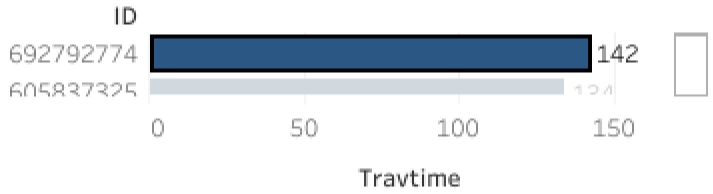


$8,950

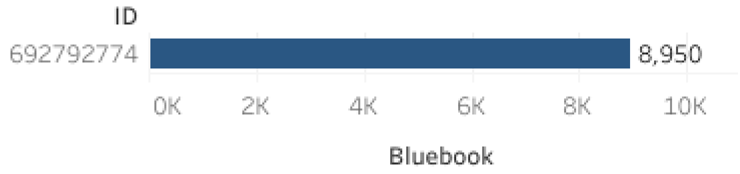
ID against Highest Income



ID against Highest TravelTime



ID against top Valued Vehicle



Summary

Demographics and Claims:

SUVs dominate claims across old and new categories. Women are the highest claimants for SUVs, while men lead for minivans and pickups.

Drivers aged 55-74 contribute the highest total claims, with senior drivers dominating specific age brackets. Urban areas contribute **93.84% of claims**, indicating a strong urban bias.

Top Claimants and Yearly Trends:

Notable years of high claim frequencies vary across vehicle types, e.g., 2002 and 2023 for SUVs, and 2001 for sports cars. High-value claims are concentrated among specific user profiles, e.g., elderly female drivers or middle-aged professionals.

Education and Profession:

Drivers with **high school or bachelor's degrees** show the highest claim frequencies, while PhD holders file the least claims. Blue-collar workers lead claims by profession, while doctors contribute the lowest totals.

Recommendations

Policy Development:

Introduce risk-adjusted premiums for SUVs and senior drivers based on claim histories. Offer safer vehicle incentives or discounts for less vulnerable car types (e.g., panel trucks).

Risk Mitigation Programs:

Launch safe driving workshops targeting high-risk demographics, such as blue-collar workers and high-frequency claimants.

Develop loyalty rewards for long-term policyholders with consistent low claim histories.

Dynamic Pricing Models:

Integrate urban-specific pricing strategies to manage the disparity between urban and rural claims.

Offer lower premiums for drivers with advanced education levels or low-risk professions.

Conclusion

The analysis underscores the importance of tailored policies and risk management strategies. By leveraging these insights, the company can enhance customer satisfaction, reduce claim frequency, and maintain long-term profitability. The proposed recommendations align with the objectives of optimizing financial reserves and developing innovative, data-driven policies.